LAB 06

Strategy Prototype: TradingView Strategy Tester, Paper Trading from the Chart and Pipeline Reconciliation

Item	Detail
Topic	Topic 2: Technical Indicators, Trading Strategies and Metrics with TradingView and Alpaca
Pipeline stage	Strategy prototype
TSC reference	A2, A3, K3, K7
Duration	30 min
Simulator page	http://localhost:8000/#/backtest?strategy=sma_cross&symbols=SPY&fast=50&slow=200&costs=alpaca_retail
Mode	No coding. Runs ONLINE with your Alpaca paper keys (Integrations page), or OFFLINE on the demo data and the built-in paper broker. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

A portfolio manager prototypes a golden-cross rule for the desk's SPY sleeve in TradingView. Before it can enter the automated pipeline, the Strategy Tester result (with commission and slippage) must be reproduced by an independent pipeline backtest, and the trader must be able to put the idea into action on a paper account straight from the chart.

Objective

Run the same MA-crossover in TradingView's Strategy Tester and in the app, reconcile the results, and try TradingView Paper Trading.

Before You Start

- Make sure the Trading Simulator is running and open http://localhost:8000 > Course Labs > Lab 06. It is installed once - see the guide 'Download, Install and Start the Trading Simulator Software' in the Learner Guide.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

Lab Steps - In the Trading Simulator

No coding is needed. Make sure the Trading Simulator is running (installed once - see the guide 'Download, Install and Start the Trading Simulator Software') and open http://localhost:8000. In the sidebar open Course Labs > Lab 06: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.

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Step 1. Backtest the SMA 50/200 crossover in the app

Run the preset backtest: SMA crossover 50/200 on SPY, Alpaca-style costs, next-bar-open fills.

Open: http://localhost:8000/#/backtest?strategy=sma_cross&symbols=SPY&fast=50&slow=200&costs=alpaca_retail (Backtest SPY SMA 50/200, Alpaca costs)

Check: About 5 round trips 2018-2025; note the net CAGR and Sharpe.

Step 2. Run it in TradingView

Export the SMA crossover Pine Script and add it to an AMEX:SPY daily chart. Open the Strategy Tester.

Open: http://localhost:8000/#/strategies?key=sma_cross (Strategies > SMA crossover > Pine)

Check: The Strategy Tester list of trades has the same entry dates (TradingView also fills on the next bar's open).

Step 3. Reconcile and paper trade on TradingView

Explain any differences (adjusted vs raw data, commission per order, slippage ticks). Optionally connect TradingView Paper Trading and place one order from the chart.

Open: http://localhost:8000/#/backtest?strategy=sma_cross&symbols=SPY&fast=50&slow=200&costs=alpaca_retail

Check: You can list at least two reasons the two backtests differ.

Expected in the app: Trade counts and net profit that agree within the expected differences (fills, costs, data).

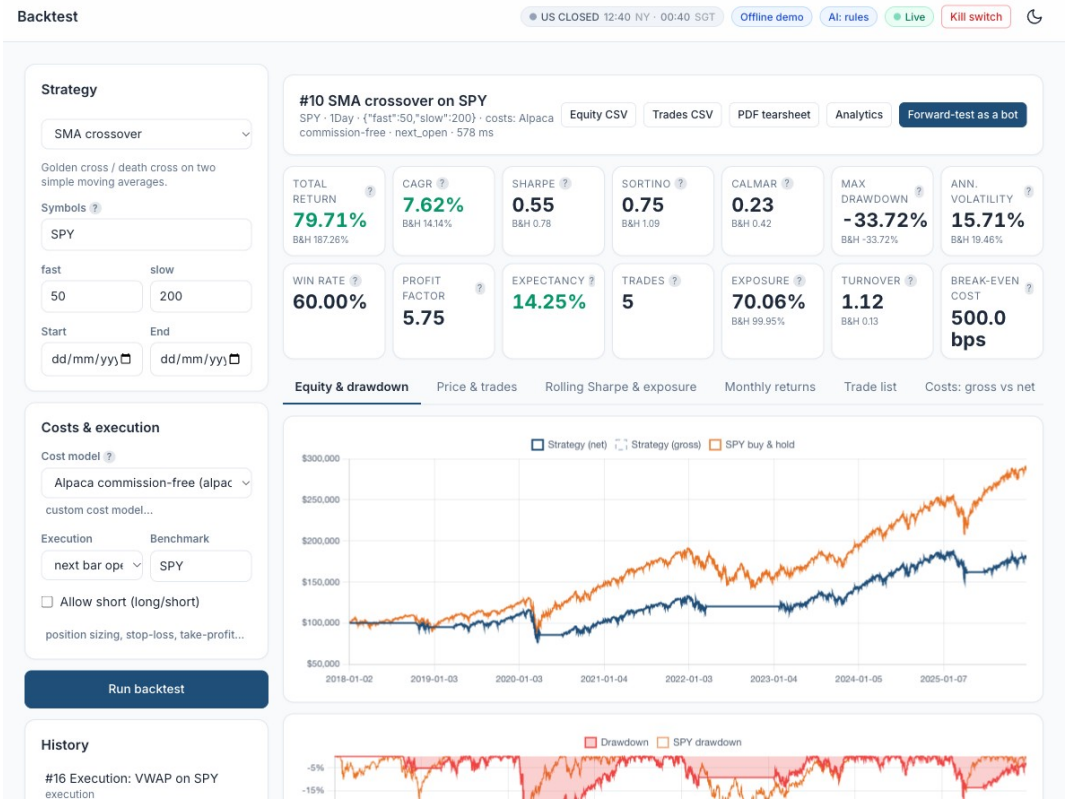


Figure: Lab 06 in the Trading Simulator.

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Optional: Let AI Write the Code

This lab needs no coding. To see, run or customise the code behind it, click a link: the AI opens with the prompt filled in and writes the code for you. All prompts are also in labs_lowcode/Prompt_Library.html and .pdf.

A crossover backtest to compare with TradingView's Strategy Tester - run it in Google Colab
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: backtest an SMA 50/200 crossover on SPY 2018-2025 (long when SMA50 > SMA200), trade at the next day's open with a 0.05% cost per trade. Print the list of trades (entry and exit dates and prices), the number of trades, net profit and maximum drawdown. Then write the same strategy in TradingView Pine Script v6 using strategy() so I can compare it in the Strategy Tester.

Open in Claude | Open in ChatGPT | Open Google Colab

Guide: Paper Trading on TradingView, Step by Step

TradingView's built-in Paper Trading account lets you place simulated orders from the chart. You use it in Lab 6 to put the moving-average strategy into action.

1. Connect Paper Trading

On Supercharts click Trade (top right). Choose Paper Trading from the broker list and click Connect. The Trading Panel opens at the bottom of the chart.

2. Set your starting capital and commission

In the Trading Panel open the account settings and set the starting balance and a commission that matches the course cost assumptions, so paper results are comparable with the Python backtest.

3. Open the order ticket

Click Buy or Sell (or Trade) to open the order ticket. Choose the side, quantity and order type - Market, Limit or Stop - and the price for a limit or stop order.

4. Add a stop loss and take profit

In the order ticket's risk settings tick Stop Loss and Take Profit and enter the levels (for example 2 x ATR below and above the entry). Then place the order.

5. Watch the order and position

Use the Trading Panel tabs: Positions (open trades and unrealised P&L), Orders (working, filled, cancelled) and Account History. Orders and positions also appear as lines on the chart, which you can drag to modify.

6. Put the strategy into action

With the Lab 6 Pine strategy on the chart, read its latest signal in the Strategy Tester's List of Trades and place the matching paper order. Record the fill price and compare it with the strategy's simulated fill.

7. Close and reset

Close a position from the Positions tab (or the x on its chart line). To start again, reset the paper account from the Trading Panel's account settings - this clears its orders and history.

Paper fills use TradingView's prices but cannot show your own market impact - compare them with the backtest's assumed costs.

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Debrief

Choosing a rule and its parameters is a modelling decision (A2); the Strategy Tester and TradingView Paper Trading are analytics and execution platforms (A3) that report standard metrics (K3). An independent reconciliation is the gate between a chart prototype and the automated pipeline, and costs typed into strategy() (K7) barely touch a slow rule but compound for a fast one.

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